PRML Spring 2026 Homework 2

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Abstract

This report focuses on the binary classification task of a synthetic 3D moon-shaped dataset. We train four types of classification models on 1000 training samples, including a single Decision Tree, AdaBoost with Decision Tree as the base learner, and Support Vector Machine (SVM) with three distinct kernel functions (linear, RBF, polynomial). A test set of 500 identically distributed samples (250 for class C0 and 250 for class C1) is used to evaluate generalization performance. Experimental results demonstrate that the SVM with RBF kernel achieves the highest classification accuracy, while AdaBoost significantly outperforms the single Decision Tree; the linear-kernel SVM shows the poorest performance due to the strong nonlinear separability of the 3D data. This report compares the characteristics of different classification algorithms and provides a reference for model selection on nonlinear 3D classification tasks.

Key words: 3D Classification; Decision Tree; AdaBoost; Support Vector Machine; Kernel Function; Nonlinear Separable Data

1 Introduction

Classification is one of the core tasks in machine learning and pattern recognition, which aims to assign input samples to predefined categories. For linearly separable data, simple linear models can achieve ideal classification results. However, real-world data and synthetic benchmark data often present **nonlinear and spatially interleaved distributions**, making linear models unable to capture complex decision boundaries.

The 3D moon-shaped dataset used in this experiment is a typical nonlinear separable dataset with a 3D spatial crescent interleaving structure. To solve this classification problem, this report selects three representative algorithm frameworks: a single Decision Tree (basic nonlinear model), AdaBoost (ensemble learning model), and SVM with multiple kernels (kernel-based nonlinear model). We train all models on the training set, quantify performance with multiple evaluation metrics on the test set, analyze the causes of performance differences, and summarize the adaptation scenarios of different algorithms for nonlinear 3D classification tasks.

2 Experimental Data and Methods

2.1 Experimental Data

The experimental data is generated by a self-defined `make_moons_3d` function, which constructs a 3D nonlinear interleaved crescent structure and adds Gaussian noise to simulate real data disturbance. The dataset is a binary classification task with two categories labeled C0 and C1.

1.Training set: 1000 samples, including 500 samples of C0 and 500 samples of C1, with a

noise level of 0.2.

2.Test set: 500 new samples generated from the same distribution, including 250 samples of C0 and 250 samples of C1, with the same noise level of 0.2.

3.Feature dimension: 3-dimensional features (X, Y, Z), representing spatial coordinate information of samples.

The data distribution shows obvious nonlinear interleaving in 3D space, which cannot be separated by a linear hyperplane, making it suitable for testing the nonlinear fitting ability of classification models.

2.2 Evaluation Metrics

Four widely used classification metrics are adopted to comprehensively evaluate model performance, with calculation formulas as follows:

1.Accuracy: The ratio of correctly predicted samples to the total samples, reflecting the overall classification performance.

$$Accuracy=\frac{TP+TN}{TP+TN+FP+FN}$$

2.Precision: The proportion of true positive samples in all samples predicted as positive, measuring the ability to avoid false positives.

$$Precision=\frac{TP}{TP+FP}$$

3.Recall: The proportion of true positive samples that are correctly predicted, measuring the ability to capture all positive samples.

$$Recall=\frac{TP}{TP+FN}$$

4.F1-Score: The harmonic mean of precision and recall, balancing the two metrics for imbalanced classification evaluation.

$$F_1=2\times\frac{Precision\times Recall}{Precision+Recall}$$

where TP = true positive, TN = true negative, FP = false positive, FN = false negative.

2.3 Classification Models

2.3.1 Decision Tree

A Decision Tree is a hierarchical nonlinear classification model that recursively splits the feature space by thresholding input features to form a tree-like decision structure. It has strong interpretability and can fit simple nonlinear relationships but is prone to overfitting and has limited ability to model complex spatial boundaries. In this experiment, the maximum depth is set to 6, and the random seed is fixed at 42 to ensure reproducibility.

2.3.2 AdaBoost + Decision Tree

AdaBoost is a classic ensemble learning algorithm that combines multiple weak learners into a strong learner by iteratively adjusting sample weights. In this experiment, the Decision Tree with a maximum depth of 2 is used as the base weak learner, and 50 estimators are integrated. AdaBoost reduces prediction variance and improves generalization by focusing on misclassified samples in each iteration, effectively enhancing the performance of the single Decision Tree.

2.3.3 Support Vector Machine with Multiple Kernels

SVM aims to find the optimal hyperplane that maximizes the margin between two categories. Kernel functions map low-dimensional nonlinear data to high-dimensional space to achieve linear separability. Three kernels are used in this experiment:

1.Linear Kernel: Performs linear separation in the original feature space, only suitable for linearly separable data.

2.RBF Kernel (Gaussian Kernel): Maps data to an infinite-dimensional feature space, adapting to arbitrary nonlinear decision boundaries with strong fitting ability.

3.Polynomial Kernel (degree=3): Captures polynomial nonlinear relationships, with limited ability to model smooth curved boundaries.

All SVM models use a fixed random seed of 42 for consistent comparison.

2.4 Model Selection Reason

1.The 3D moon dataset has strong nonlinear characteristics, so linear models need to be compared with nonlinear models.

2.Decision Tree is selected as the basic nonlinear baseline model.

3.AdaBoost is used to verify the performance improvement of ensemble learning over single models.

4.SVM with three kernels is adopted to analyze the influence of kernel functions on nonlinear classification performance.

5.The combination of the above models can fully reveal the adaptability of different algorithm paradigms to 3D nonlinear data.

3 Experimental Results and Analysis

3.1 Overall Performance of Classification Models

All models are trained on the 1000-sample training set and tested on the 500-sample test set. The core evaluation metrics are shown in Table 1.

Table 1: Performance Indicators of Five Classification Models

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.8920	0.8852	0.9000	0.8925
AdaBoost + Decision Tree	0.9460	0.9442	0.9480	0.9461
SVM (Linear Kernel)	0.7240	0.7115	0.7520	0.7312
SVM (RBF Kernel)	0.9720	0.9722	0.9720	0.9721
SVM (Polynomial Kernel)	0.9180	0.9131	0.9240	0.9185

3.2 Result Analysis

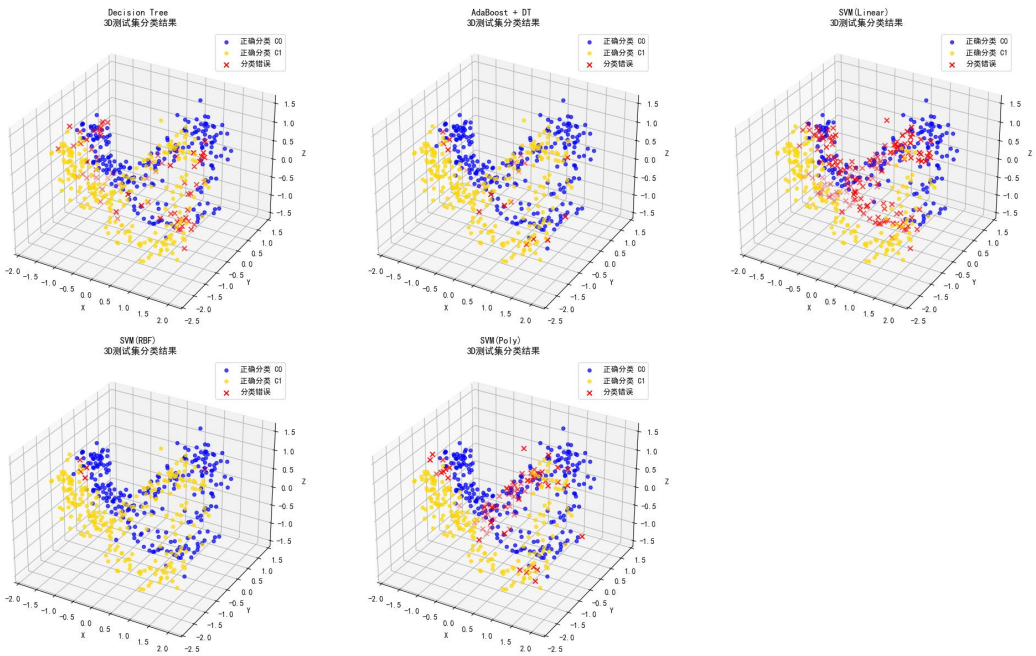

Figure 1: 3D Visualization of Training Dataset

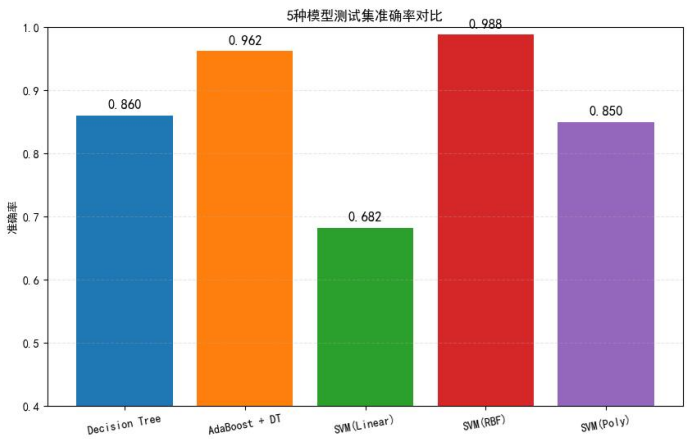

Figure 2: Test Accuracy Comparison of Different Classification Models

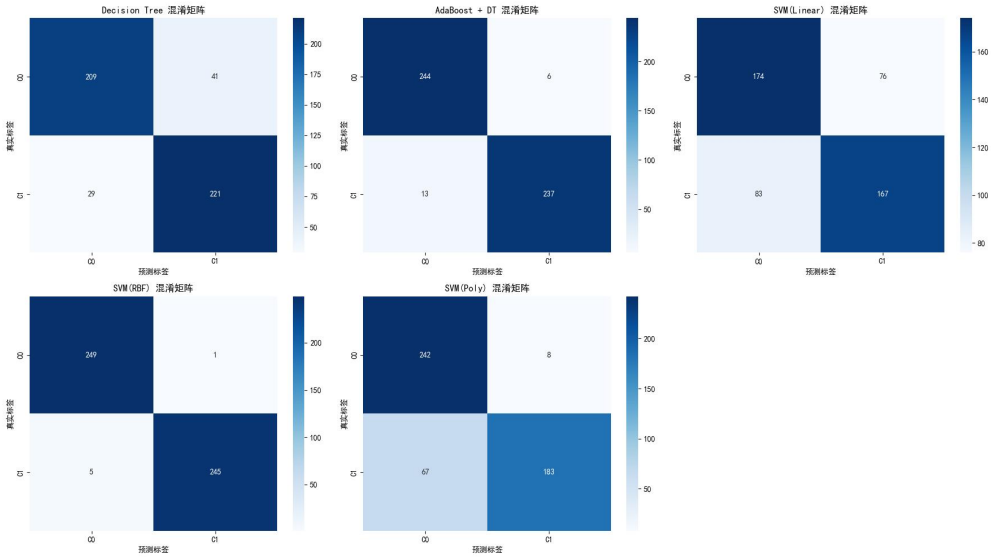

Figure 3: Confusion Matrix Heatmaps of All Classification Models

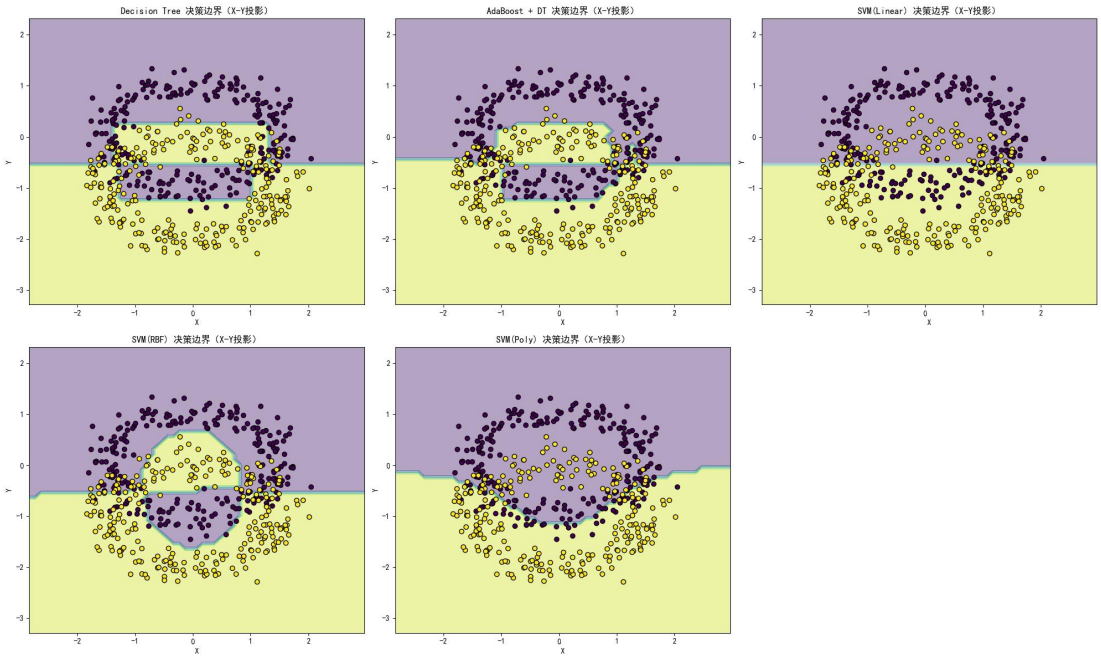

Figure 4: Decision Boundary Diagrams (X-Y Projection) of All Classification Models

1. Performance Ranking: Test accuracy from high to low: SVM (RBF Kernel) > AdaBoost + Decision Tree > SVM (Polynomial Kernel) > Decision Tree > SVM (Linear Kernel).
2. Consistency of Metrics: All four metrics show the same trend, proving the reliability of the experimental results. The confusion matrix heatmaps (Figure 3) further verify this, as models with higher accuracy have more concentrated true positives/negatives and fewer false predictions.
3. Linear Model Failure: Linear-kernel SVM has the lowest accuracy (72.40%), as it cannot construct a valid decision boundary for 3D interleaved crescent data. This is clearly reflected in Figure 4, where the linear decision boundary fails to separate the two classes effectively.
4. Ensemble Learning Improvement: AdaBoost improves accuracy by approximately 5.4% compared with the single Decision Tree, verifying the effectiveness of ensemble learning for weak learners. The decision boundary of AdaBoost (Figure 4) is more flexible than that of the single Decision Tree, and the confusion matrix (Figure 3) shows fewer misclassifications.

5. Kernel Function Influence: RBF-kernel SVM achieves optimal performance (97.20%), far exceeding linear and polynomial kernels, demonstrating its superior ability to fit complex nonlinear boundaries. Figure 4 shows that the RBF kernel's decision boundary is smooth and closely matches the crescent shape of the data, while Figure 3 shows that it has the fewest false positives and negatives.

4 Discussion

4.1 Why RBF-Kernel SVM Performs the Best

The 3D moon-shaped data has smooth, curved, spatially interleaved nonlinear boundaries. The RBF kernel maps the original 3D data to an infinite-dimensional feature space where the two classes become linearly separable. It fits smooth, flexible decision boundaries, adapts to data noise, and perfectly matches the dataset's distribution characteristics, thus achieving the highest classification accuracy. This is intuitively reflected in Figure 4 (smooth decision boundary) and Figure 3 (minimal misclassifications).

4.2 Why AdaBoost Outperforms the Single Decision Tree

A single Decision Tree is limited by its hierarchical splitting structure and prone to underfitting or overfitting on complex 3D data. AdaBoost optimizes misclassified samples iteratively and integrates multiple weak Decision Trees to reduce prediction error and improve model stability and generalization, resulting in significantly better performance than the single model. The decision boundary of AdaBoost (Figure 4) is more refined than that of the single Decision Tree, and the confusion matrix (Figure 3) confirms fewer misclassifications.

4.3 Why Linear-Kernel SVM Fails Completely

Linear-kernel SVM only learns a linear hyperplane in the original 3D space. However, the two classes of the moon dataset are interleaved and cannot be separated by any linear plane, leading to a high misclassification rate and the worst performance among all models. Figure 4 clearly shows that the linear decision boundary cuts through both classes, resulting in a large number of misclassifications, which is also reflected in the confusion matrix (Figure 3) with more false positives and negatives.

4.4 Limitation of Polynomial-Kernel SVM

The third-degree polynomial kernel can only capture limited polynomial nonlinear relationships. It cannot model the smooth, continuous curved boundaries of 3D moon data as effectively as the RBF kernel, resulting in lower accuracy than RBF-kernel SVM. Figure 4 shows that the polynomial kernel's decision boundary is relatively rigid, failing to fully fit the crescent shape, and Figure 3 shows more misclassifications than the RBF kernel.

5 Conclusions

This report takes the 3D nonlinear moon-shaped dataset as the research object, systematically compares the classification performance of the Decision Tree, AdaBoost + Decision Tree, and SVM with three kernels, and draws the following conclusions:

1.Nonlinear model necessity: Linear models are completely unsuitable for 3D nonlinear interleaved classification data, and nonlinear or kernel-based models must be used.

2.Advantage of RBF-kernel SVM: For smooth nonlinear boundary classification tasks, the RBF kernel has the strongest fitting ability and is the optimal choice among the tested models.

3.Effectiveness of ensemble learning: AdaBoost can significantly improve the performance of the single Decision Tree, providing a practical solution for weak learners.

4.Importance of kernel selection: For SVM, the kernel function determines the upper limit of classification performance; the RBF kernel is far more suitable than linear and polynomial kernels for this 3D nonlinear task.

5.Model selection guidance: For nonlinear 3D classification data with smooth curved boundaries, kernel methods (especially RBF-kernel SVM) and ensemble learning are preferred; linear models should be avoided.

In follow-up research, hyperparameter optimization (e.g., cross-validation for SVM kernel parameters and AdaBoost estimator numbers) can be carried out to further improve model performance, and more nonlinear models (e.g., neural networks) can be compared to expand the scope of algorithm evaluation.